Jioh In

Ann Arbor, MI | jiohin@umich.edu | linkedin.com/in/jioh-in | github.com/retz8 | retz8.github.io

EDUCATION

University of Michigan - Ann Arbor **Expected Dec 2028**

B.S.E. in Computer Science *GPA: 3.91/4.00*

- On military service leave (Republic of Korea Army), Oct 2025 - Apr 2027; returning to coursework Fall 2027
- Relevant Courseworks: Data Structures and Algorithms · Computer Organization · Web Systems

WORK EXPERIENCE

Tesla Inc. · **Software Engineer Intern** · **Fremont, CA** **May 2025 - Aug 2025**

- Primary frontend engineer on team of nine; built tesla.com/energy-savings from scratch in Next.js, TypeScript, and SCSS, calculator letting customers estimate utility bill savings for their own system before ordering energy products
- Owned order workflow at tesla.com/energy/design through Puerto Rico market launch and backend API migration; built Next.js BFF isolating frontend from changing price calculation endpoints, added i18n and internal PDF export
- Ran A/B tests on mega-menu navigation and new recommendation logic against existing, driving rollout decisions
- Automated UI regression suite for order workflow with Pytest and Selenium, integrated into Jenkins CI/CD

UM Transportation Research Institute · **Research Assistant** · **Ann Arbor, MI** **Jan 2025 - Apr 2025**

- Devised 3D simulation tool of human and wheelchair models in Three.js and WebGL, fitting posture to wheelchair parameters with PCA and linear regression; shipped as public research tool for U.S. wheelchair seating standards

UMich KISA Dev Team · **Founder & Engineering Lead** · **Ann Arbor, MI** **Nov 2023 - Present**

- Founded organization's first engineering team, growing it to 10+ contributors and leading 7 engineers at peak
- Led six production projects end to end: admin tooling, event analytics, and job curation platform on API partnership with Wanted, Korea's top recruiting platform; scoped tasks to member skill level, reviewed 125+ PRs

PROJECTS

Generative UI on A2UI · a2ui-project/a2ui, A2UI GitHub, A2UIVerse **Jul 2026 - Present**

- Contributor to a2ui-project/a2ui, Google's open source Generative UI protocol, with merged renderer and spec fixes
- Built GitHub client where agent generates UI at runtime over A2UI instead of hardcoded screens; ported 56 Primer components as A2UI catalog, wired agent to GitHub MCP server for live data, added replayable, branchable history
- Building A2UIVerse, where LLM orchestrator routes requests across installed A2A agents into one merged screen

Pastiche · open-source Claude Code Plugin **May 2026 - Jun 2026**

- Built plugin constraining agent-generated UI to team's component library, extending Google's DESIGN.md with auto-extracted component catalog and curated scenario mappings so agents ship mergeable code, not prototypes
- Wrote supporting toolchain in TypeScript and Rust: AST extractor on ts-morph deriving component and token catalog from source without LLM inference, and CLI linter validating schema and drift across contract documents

UMich KISA Design System · npm: @umichkisa-ds/web, @umichkisa-ds/form **Mar 2026 - May 2026**

- Designed and published production design system from scratch; orchestrated builds and type generation across workspaces with pnpm + Turborepo, automating publication via GitHub Actions across 33 releases
- Shipped 41 TypeScript components on Radix UI primitives and Tailwind v4, typed form layer over react-hook-form, and docs site on two-tier OKLCH token architecture; component and form layers covered by Vitest unit tests
- Migrated organization's live production website onto design system via phased rollout in four weeks, zero downtime

Snippet Newsletter · iris-codes.com/snippet **Jan 2026 - Present**

- Publish twice-weekly code-reading newsletter on premise that understanding code becomes bottleneck as AI writes more of it; 160+ snippets in 3 languages since Feb 2026 on autonomous n8n and Claude Code skills pipeline

UMich KISA Web Platform · umichkisa.com, umichkisa.com/pocha **Nov 2023 - Present**

- Built and maintain official website of student association, delivering announcements, bulletin board, and event tools to 500+ member Korean community on Next.js and Flask, deployed on Vercel and AWS Elastic Beanstalk
- Engineered real-time ordering platform for pop-up events, scaling Flask backend to 200+ concurrent users and 5,000+ orders; integrated Stripe API for 600+ live transactions, cut manual handling 50% with Socket.io tracking

SKILLS

- **Programming Languages:** Python, C, C++, Java, TypeScript, JavaScript, Rust, SQL, Bash/Shell, UNIX/Linux
- **Frameworks:** React, Next.js, React Native, Node.js, Flask, Tailwind/SCSS, Docker, Jenkins, AWS, REST, GraphQL